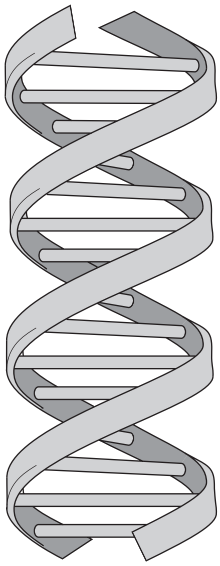
Examiner only

8. DNA is made of two long chains of alternating molecules of sugar and phosphate connected by bases. The chains are twisted to form a double helix as shown in the diagram.



(a) A sample of DNA was analysed and found to contain 27% thymine.

(i) Name the complementary base to thymine. [1]

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(ii) Calculate the percentage of **cytosine** that would be present in this sample. [2]

Cytosine = ... %

Examiner only

(b) (i) Explain how the sequence of bases on DNA determines the structure of a protein. [3]

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The sequence of bases on a small section of DNA is shown below.

TGA CTG GTC AAC

This section of DNA can undergo two different mutations.

Mutation 1 **TGA CAG CTC AAC**

Mutation 2 **TGA CTG GTC ATC**

(ii) Using the information and your own knowledge conclude why mutation 1 may affect an organism more than mutation 2. [3]

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END OF PAPER